Saurav Rauthan

AI Engineer

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SUMMARY

AI/ML engineer (B.Tech, 2026) with hands-on project experience in multi-agent LLM applications, computer vision, deep learning, and full-stack deployment (FastAPI, Supabase, Render, Vercel, Streamlit). Comfortable across the Generative AI stack — LLMs, RAG, LangChain, vector databases, and prompt engineering — with a strong Python foundation and a bias toward shipping working, production-style systems rather than notebooks. Fast learner, self-driven, and available to start immediately on remote agentic-AI product work.

EDUCATION

Bachelor of Technology in Artificial Intelligence and Machine Learning, *Dr. APJ Abdul Kalam Institute of Technology, Champawat* 08/2023 – 07/2026

- Relevant Coursework: Data Structures, Algorithm Analysis, AI, Machine Learning, Operating Systems, Computer Networks, Computer Architecture, DBMS

EXPERIENCE

Data Science Intern (Virtual), *CodSoft* 06/2024 – 08/2024

- Completed 5 hands-on data science tasks across classification and recommendation problems, including Titanic survival prediction, customer churn prediction, and a Netflix-style movie recommendation system.
- Handled each task end to end in Python, from data cleaning and exploration to model building and evaluation.

PROJECTS

TravelCraft — Multi-Agent Travel Planner | Live Demo

- Built a full-stack travel planner where a Groq-powered Master Agent parses natural-language requests and coordinates 4 specialized agents (Transportation, Hotel, Budget, Weather), integrating SerpAPI (live flights, hotels) and Open-Meteo (weather) and asking follow-up questions when trip details are missing.
- Streamed real-time agent progress to the frontend over WebSockets during 10+ second multi-agent research runs; implemented JWT authentication with Redis-backed login handling and API rate limiting, plus user-specific history in Supabase (PostgreSQL) and saved travel preferences.
- Integrated the PayPal Orders API (Sandbox) with backend-verified order creation and capture instead of trusting frontend approval.
- Deployed as a 3-tier app: frontend on Vercel, FastAPI backend on Render, and Supabase (PostgreSQL) via its session pooler, with production CORS and secured environment variables.

AI-Powered Resume-JD Matcher (ATS Scorer) | Live Demo

- Built a full-stack web app scoring resume-JD fit via semantic similarity (Sentence Transformers) instead of plain keyword matching, with a 4-category breakdown (formatting, keywords, content, ATS compatibility); validated on 7 real resumes and recalibrated so a deliberately weak one dropped from 62 to 34.
- Integrated the Groq API (Llama 3) for actionable, LLM-written improvement suggestions on top of the NLP score.
- Implemented Supabase auth (email + Google OAuth) so users can save and revisit past analyses, and added PDF report export via WeasyPrint + Jinja2.

AI-Powered Attendance System | Live Demo

- Built an end-to-end attendance portal automating student attendance via real-time facial recognition; correctly identified all 10+ registered users in testing.
- Integrated a Supabase (PostgreSQL) backend for authentication and real-time attendance logging, and engineered a dual-panel interface for registration and record tracking.
- Optimized face-matching latency for fast, multi-face scanning — the kind of inference-cost/latency trade-off central to production AI features.

ACHIEVEMENTS / EXTRACURRICULAR

- Scored 93.60 percentile in Naukri Campus Young Turks 2026 (HCLTech presents), India's largest skill contest.
- Secured 2nd position at UCOST (Uttarakhand State Council for Science and Technology) for presenting an innovative tech-driven project at the state level.

SKILLS

Generative AI & NLP	LLMs, Agentic AI, Multi-Agent Systems, RAG, LangChain, Hugging Face Transformers, Sentence Transformers, Semantic Search, Prompt Engineering, Llama 3, Groq/OpenAI/Gemini APIs, Vector Databases (ChromaDB)
ML & Deep Learning	Machine Learning, Deep Learning, Neural Networks, CNNs, GANs, Computer Vision, PyTorch, Scikit-Learn, OpenCV, MediaPipe
Languages	Python, SQL, JavaScript, HTML/CSS, R
Data, CS & Tools	NumPy, Pandas, Matplotlib, DSA, Object-Oriented Programming, Git/GitHub, VS Code, Jupyter Notebook, Google Colab
Backend & APIs	API Development, FastAPI, Flask, REST APIs, WebSockets, JWT Authentication, PostgreSQL (Supabase), Redis, PayPal API, SerpAPI, Open-Meteo
Deployment & DevOps	Model Deployment, AWS, Docker, Kubernetes, CI/CD (GitHub Actions), Render, Vercel, Streamlit